

Limit Points of a Sequence :

A real number l is said to be a limit point of a sequence $\langle a_n \rangle$ if every neighbourhood of l contains infinitely many terms of sequence.

Thus $l \in \mathbb{R}$ is a limit point of seq. $\langle a_n \rangle$

$\Leftrightarrow$ Every neighbourhood of l contains infinitely many terms of the sequence.

$\Leftrightarrow \forall \epsilon > 0, a_n \in (l - \epsilon, l + \epsilon)$ for infinitely many values of n .

$\Leftrightarrow \forall \epsilon > 0, |a_n - l| < \epsilon$ for infinitely many values of n .

Notes: Limit point of a sequence is different from limit of a sequence.

2. $\exists$ If $a_n = l$ for infinitely many value of n then l is a limit point of $\langle a_n \rangle$.

3. $\exists$ If for an $\epsilon > 0, a_n \in (l - \epsilon, l + \epsilon)$ for finitely many values of n , then l can't be a limit point of $\langle a_n \rangle$.

Examples.

① 0 is the limit point of seq. $\langle \frac{1}{n} \rangle$
because every nbhd of 0 contains infinitely many terms of seq. $\langle \frac{1}{n} \rangle$.

② Sequence $\langle (-1)^n \rangle$ has two limit points -1 and 1

$$\text{Let } a_n = (-1)^n = \begin{cases} -1 & n \text{ is odd} \\ 1 & n \text{ is even} \end{cases}$$

Since every nbhd of -1 contains all the odd terms of $\text{seq. } \langle (-1)^n \rangle$

$\Rightarrow -1$ is a limit point.

Similarly, Every nbhd of 1 contains all the even terms of $\text{seq. } \langle (-1)^n \rangle$.

$\Rightarrow 1$ is a limit point.

③ The sequence $\langle n \rangle$ has no limit point.

Let l be any real number, then the nbhd $(l - \frac{1}{4}, l + \frac{1}{4})$ of l contains at most one term of $\text{seq. } \langle n \rangle$.

$\Rightarrow l$ is not a limit point of the $\text{seq. } \langle n \rangle$.

Theorem:- If l is a limit point of the range of $\text{seq. } \langle a_n \rangle$, then l is a limit point of $\text{seq. } \langle a_n \rangle$.

Converse of the above theorem is not true.

Ex
$$a_n = 1 + (-1)^n = \begin{cases} 0 & \text{when } n \text{ is odd} \\ 2 & \text{when } n \text{ is even} \end{cases}$$

Clearly $0, 2$ are the limit points of $\text{seq. } \langle a_n \rangle$.

Range set of $\text{seq. } \langle a_n \rangle = \{0, 2\}$ is a finite set.

Since finite set has no limit point.

$\Rightarrow$ Range set has no limit points.

Theorem: (The sufficient condition for the existence of a limit point of a sequence.)

Bolzano-Weierstrass Theorem for Sequences

Every bounded sequence has at least one limit point.

Theorem: Every bounded sequence has the greatest and the least limit points.

Limit Superior and Limit Inferior of a sequence:

Let $\langle a_n \rangle$ be a bounded sequence, then the sequence has the least and greatest limit

The least limit point of $\langle a_n \rangle$ is called the limit inferior of $\langle a_n \rangle$ and denoted by

$$\liminf_{n \rightarrow \infty} a_n \text{ or } \underline{\lim}_{n \rightarrow \infty} a_n$$

The greatest limit point of $\langle a_n \rangle$ is called the limit superior of $\langle a_n \rangle$ and denoted by

$$\limsup_{n \rightarrow \infty} a_n \text{ or } \overline{\lim}_{n \rightarrow \infty} a_n$$

Note 1: If $\langle a_n \rangle$ is unbounded above, we write

$$\limsup_{n \rightarrow \infty} a_n = \infty$$

If $\langle a_n \rangle$ is unbounded below, we write

$$\liminf_{n \rightarrow \infty} a_n = -\infty$$

Note 2:

$$\limsup_{n \rightarrow \infty} a_n \geq \liminf_{n \rightarrow \infty} a_n$$

Examples:

- (i) For the seq. $\langle a_n \rangle$ defined by $a_n = (-1)^n \forall n \in \mathbb{N}$,
the only limit points are -1 and 1

The set of limit points $= \{-1, 1\}$ which is finite
so bounded

$$\therefore \liminf_{n \rightarrow \infty} a_n = -1$$

$$\limsup_{n \rightarrow \infty} a_n = 1$$

- (ii) For the seq. $\langle a_n \rangle$ defined by $a_n = \frac{1}{n}, \forall n \in \mathbb{N}$, the
only limit points ~~are~~ is 0 .

The set of limit points $= \{0\}$ is finite so bounded

$$\Rightarrow \liminf_{n \rightarrow \infty} a_n = 0 = \limsup_{n \rightarrow \infty} a_n$$

- (iii) For the constant sequence $a_n = k \forall n \in \mathbb{N}$

$$\limsup_{n \rightarrow \infty} a_n = k = \liminf_{n \rightarrow \infty} a_n$$

- (iv) $\forall n \quad a_n = \begin{cases} 2 & \text{when } n \text{ is odd} \\ -n & \text{when } n \text{ is even} \end{cases}$

Set of $a_n = \{2, -2, 2, -4, 2, -6, 2, -8, \dots\}$

set of limit point $= \{2\}$

$\liminf_{n \rightarrow \infty} a_n = -\infty$ and $\limsup_{n \rightarrow \infty} a_n = 2$

It is
not bounded
above seq.
only
not bounded

(v) let $a_n = (-1)^n \cdot n \quad \forall n \in \mathbb{N}$

$$\text{so } a_n = \{-1, 2, -3, 4, -5, 6, -7, 8, \dots\}$$

$$\liminf_{n \rightarrow \infty} a_n = -\infty$$

$$\limsup_{n \rightarrow \infty} a_n = \infty$$

Theorem: A seq. $\langle a_n \rangle$ converges to l
iff

$$\limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = l.$$

Theorem: (i) If a sequence $\langle a_n \rangle$ is such that

$$\limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = +\infty$$

then $\langle a_n \rangle$ diverges to ∞

(ii) If a sequence $\langle a_n \rangle$ is such that

$$\limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = -\infty$$

then $\langle a_n \rangle$ diverges to $-\infty$.

Ques) Find the limit point for the following sequences:

(i) $\langle 5 \rangle$ (ii) $\langle \frac{(-1)^n}{n} \rangle$ (iii) $\langle n \rangle$ (iv) $\langle (-1)^n \left(1 + \frac{1}{n}\right) \rangle$ (v) $\langle \left(1 + \frac{1}{n}\right)^{n+1} \rangle$

Soln (i) $\langle a_n \rangle = \langle 5 \rangle = \langle 5, 5, 5, \dots \rangle$

It is constant sequence converging to 5.
It has only one limit point 5.

(ii) $\langle a_n \rangle = \langle \frac{(-1)^n}{n} \rangle = \begin{cases} -\frac{1}{n} & \text{when } n \text{ is odd} \\ \frac{1}{n} & \text{when } n \text{ is even} \end{cases}$

$\lim_{n \rightarrow \infty} a_n = 0 \Rightarrow a_n \rightarrow 0$

So $\langle a_n \rangle$ has only one limit point 0.

(iii) $\langle a_n \rangle = \langle n \rangle = \langle 1, 2, 3, 4, \dots \rangle$

for any $l \in \mathbb{R}$, then neighbourhood $(l - \frac{1}{4}, l + \frac{1}{4})$ of l contains at most one term of seq. $\langle n \rangle$.

$\Rightarrow l$ is not a limit point of $\langle n \rangle$.

(iv) $\langle a_n \rangle = \langle (-1)^n \left(1 + \frac{1}{n}\right) \rangle = \begin{cases} -\left(1 + \frac{1}{n}\right) & \text{when } n \text{ is odd} \\ \left(1 + \frac{1}{n}\right) & \text{when } n \text{ is even} \end{cases}$

$\lim_{n \rightarrow \infty} a_n = \begin{cases} -1 & \text{when } n \text{ is odd} \\ 1 & \text{when } n \text{ is even} \end{cases}$

Limit points are -1 & 1

(v) $\langle a_n \rangle = \langle \left(1 + \frac{1}{n}\right)^{n+1} \rangle$

$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \cdot \left(1 + \frac{1}{n}\right) = e \times 1 = e$

$$\Rightarrow a_n \rightarrow e$$

The seq. has one limit point e .

Question > Find the limit superior and limit inferior of each of the following sequences:

(i) $\langle 1, 3, 5, 1, 3, 5, 1, 3, 5, \dots \rangle$

(ii) $\langle 1, 5, 17, 19, 1, 5, 17, 19, 1, 5, 17, 19, \dots \rangle$

(iii) $\langle a_n \rangle$ where $a_n = \sin \frac{n\pi}{3}$

(iv) $\langle a_n \rangle$ where $a_n = (-1)^n (2^n + 3^n)$

Soln >

(i)
$$a_n = \begin{cases} 1 & \text{if } n = 3m-2 \\ 3 & \text{if } n = 3m-1, m \in \mathbb{N} \\ 5 & \text{if } n = 3m \end{cases}$$

Set of limit point of $\langle a_n \rangle = \{1, 3, 5\}$

$$\limsup_{n \rightarrow \infty} a_n = \max \{1, 3, 5\} = 5$$

$$\liminf_{n \rightarrow \infty} a_n = \min \{1, 3, 5\} = 1$$

(ii) Let $\langle a_n \rangle = \langle 1, 5, 17, 19, 1, 5, 17, 19, \dots \rangle$

Set of limit point of $\langle a_n \rangle = \{1, 5, 17, 19\}$

~~limsup~~

$$\limsup_{n \rightarrow \infty} a_n = \max \{1, 5, 17, 19\} = 19$$

$$\liminf_{n \rightarrow \infty} a_n = \min \{1, 5, 17, 19\} = 1$$

$$(iii) \quad a_n = \sin \frac{n\pi}{3} = \begin{cases} 0 & \text{if } n=3m \\ \frac{\sqrt{3}}{2} & \text{if } n=6m-5 \text{ or } 6m-4 \\ -\frac{\sqrt{3}}{2} & \text{if } n=6m-2 \text{ or } 6m-1 \end{cases} \quad \text{where } m \in \mathbb{N}$$

$$\text{Set of limit points of } \langle a_n \rangle = \left\{ 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2} \right\}$$

$$\limsup_{n \rightarrow \infty} a_n = \max \left\{ 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2} \right\} = \frac{\sqrt{3}}{2}$$

$$\liminf_{n \rightarrow \infty} a_n = \min \left\{ 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2} \right\} = -\frac{\sqrt{3}}{2}$$

$$(iv) \quad a_n = (-1)^n (2^n + 3^n) = \begin{cases} -(2^n + 3^n) & \text{when } n \text{ is odd} \\ (2^n + 3^n) & \text{when } n \text{ is even} \end{cases}$$

$$\lim_{n \rightarrow \infty} (2^n + 3^n) = \infty \quad \text{Set of limit points of } \langle a_n \rangle = \{-\infty, \infty\}$$

$$\limsup_{n \rightarrow \infty} a_n = \infty$$

$$\liminf_{n \rightarrow \infty} a_n = -\infty$$

Subsequences:

Let $\langle a_n \rangle$ be a given sequence.

If $\langle n_k \rangle$ is a strictly increasing sequence of natural numbers (i.e. $n_1 < n_2 < n_3 < \dots$), then $\langle a_{n_k} \rangle$ is called a subsequence of $\langle a_n \rangle$.

for example:

(i) $\langle a_{2n} \rangle$, $\langle a_{2n-1} \rangle$, $\langle a_{n^2} \rangle$ are all subsequences of $\langle a_n \rangle$.

(ii) The sequences $\langle 2, 4, 6, \dots \rangle$, $\langle 1, 3, 5, \dots \rangle$, $\langle 1, 4, 9, 16, \dots \rangle$ are all subsequences of the sequence $\langle n \rangle$.

Note:

① If $\langle u_n \rangle$ is a subsequence of $\langle a_n \rangle$ and $\langle v_n \rangle$ is a subsequence of $\langle u_n \rangle$, then $\langle v_n \rangle$ is a subsequence of $\langle a_n \rangle$.

② Every sequence is a subsequence of itself.

Theorem: If a sequence $\langle a_n \rangle$ converges to l , then every subsequence of $\langle a_n \rangle$ also converges to l .

Note: Converse of the above theorem is not true.

Example $\langle a_n \rangle = \langle (-1)^n \rangle = \langle -1, 1, -1, 1, \dots \rangle$

Seq. $\langle a_n \rangle$ is not convergent because it's not having unique limit.

$$\langle a_{2n-1} \rangle = \langle a_1, a_3, a_5, \dots \rangle = \langle -1, -1, -1, \dots \rangle$$

$$\langle a_{2n} \rangle = \langle a_2, a_4, a_6, \dots \rangle = \langle 1, 1, 1, \dots \rangle$$

$\langle a_{2n-1} \rangle$ and $\langle a_{2n} \rangle$ both are subsequences of $\langle a_n \rangle$

$$\langle a_{2n} \rangle \rightarrow 1 \quad \text{and} \quad \langle a_{2n-1} \rangle \rightarrow -1$$

but $\langle a_n \rangle$ is not convergent.

Theorem: If the subsequences $\langle a_{2n-1} \rangle$ and $\langle a_{2n} \rangle$ of a seq. $\langle a_n \rangle$, both are converges to same limit l , then $\langle a_n \rangle$ also converges to l .

Theorem: (i) If $\langle a_n \rangle$ is diverges to ∞
then every subseq. of $\langle a_n \rangle$ diverges to ∞ .

(ii) If $\langle a_n \rangle$ is diverges to $-\infty$
then every subseq. of $\langle a_n \rangle$ diverges to $-\infty$.

Note:

Converse of above theorem is not true.

Example: Let $\langle a_n \rangle$ be a seq. defined as

$$a_n = (-1)^n \cdot n = \begin{cases} -n & \text{if } n \text{ is odd} \\ n & \text{if } n \text{ is even} \end{cases}$$

$$\langle a_{2n-1} \rangle = \langle -n \rangle \text{ diverges to } -\infty$$

$$\langle a_{2n} \rangle = \langle n \rangle \text{ diverges to } +\infty$$

but seq. $\langle a_n \rangle$ oscillates infinitely between $-\infty$ to $+\infty$

Theorem: Every ~~seq~~ sequence contains a monotonic subsequence.

Ex let $\langle a_n \rangle = \langle (-1)^n \cdot n \rangle = \langle -1, 2, -3, 4, -5, 6, \dots \rangle$
 Its subsequences are.

$\langle a_{2n-1} \rangle = \langle -1, -2, -3, -5, \dots \rangle$ is monotonically decreasing
 $\langle a_{2n} \rangle = \langle 2, 4, 6, \dots \rangle$ is monotonically increasing.

Bolzano - Weierstrass Theorem for Subsequences:

Every bounded sequence has convergent subsequence.

Ex. consider a bounded seq. $\langle a_n \rangle = \langle (-1)^n \rangle$.
 It has subsequences

$$\langle a_{2n} \rangle = \langle 1, 1, 1, \dots \rangle \rightarrow 1$$

$$\langle a_{2n-1} \rangle = \langle -1, -1, -1, \dots \rangle \rightarrow -1.$$